2025 JUNE PURE MATHEMATICS PAPER 1 SOLUTIONS

6042/1 ADVANCED LEVEL

PURE MATHEMATICS
PAPER 1
6042/1

3 hours

ONLINE REVISION

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INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the spaces provided.
- Answer all questions.
- Numerical answers should be given to appropriate accuracy.

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets [ ] at the end of each question.

This document contains 22 printed pages.

QUESTIONS

1. Find the series expansion of the function,

$$f(x) = \frac{3-x}{3+2x}$$

up to and including the term in x^3. State the range of values of x for which the expansion is valid. [6]

2. Solve the equation:

$$\frac{3^x(9^x)^2}{81} = 27^x.$$

[4]

3. A graph has the following parametric equations:

$$x = \frac{\sin\theta}{\cos\theta}; \quad y = 1 - \sin^2\theta; \quad \text{for } -\pi < 2\theta < \pi.$$

(a) Find $\frac{dy}{dx}$. [4]

(b) Find the coordinates of the turning point of the graph. [4]

4. Find the range of values of k for which the equation $kx^2 + 4x + k = 3$ has real roots. [5]

5. Variables x and y are related by the equation:

$$y = x^m e^n, \quad \text{where } m \text{ and } n \text{ are constants.}$$

(a) Show that this relationship can be reduced to a linear form. [2]

(b) A straight-line graph representing the relationship between x and y in (a) passes through points P and Q with coordinates $(30.5, 0.05)$ and $(0, -3)$, respectively. Find the value of m and the value of n. [3]

6. When the function $f(x) = (x-2)(x+3) + ax + b$ is divided by $x - 2$, the remainder is 8, and when divided by $x + 3$, the remainder is -2.

(a) Find the value of a and the value of b. [4]

(b) Hence or otherwise, find the remainder when $f(x)$ is divided by $(x-2)(x+3)$. [2]

7. Express y in terms of x given that:

$$x\frac{dy}{dx} = y + 1, \quad \text{and } y = 4 \text{ when } x = 2.$$

[6]

8. A chord AB of a circle of radius 8 cm subtends an angle of $\frac{\pi}{8}$ radians at the centre of the circle. Calculate the:

   (a) length of the major arc AB, [2]

   (b) area enclosed by the chord AB and the minor arc AB. [4]

9. (a) Given:

$$P = \begin{pmatrix} -1 & 1 & 3 \\ 2 & -2 & -6 \\ -4 & 4 & 12 \end{pmatrix} \text{ and } Q = \begin{pmatrix} 3 & -2 & 0 \\ -1 & 3 & 7 \\ 2 & -1 & 1 \end{pmatrix},$$

   find PQ. [3]

   (b) Show that:

$$\begin{pmatrix} -6 & -2 & 2 \\ 4 & -6 & 0 \\ 8 & 10 & -4 \end{pmatrix},$$

   does not have an inverse. [3]

10. F varies directly as the square of W, and when $W = 830$, $F = 165$.

    (a) Find the equation connecting F and W. [4]

    (b) Calculate W when $F = 132$. [2]

11. (a) Show by calculation that the equation $e^{-\frac{1}{2}x} = 3x - 1$ has a root between $x = 0$ and $x = 1$. [3]

    (b) Show that the equation $e^{-\frac{1}{2}x} = 3x - 1$ can be expressed as the iteration formula

$$x_{n+1} = -2\ln(3x_n - 1)$$

    [2]

    (c) Taking $x_1 = 0.5$ and using the iteration formula

$$x_{n+1} = -2\ln(3x_n - 1),$$

    find the value of x_2 correct to 3 decimal places. [2]

12. Find $\sqrt[3]{1 - \sqrt{3}i}$ in the form $a + bi$. [8]

13. A line l passes through the point $(-2, 3)$ and is perpendicular to the line $2x - 3y = 4$.

    (a) Find the coordinates of the point of intersection of l and the line $2x - 3y = 4$. [6]

    (b) Find the distance between the point $(-2, 3)$ and the point of intersection of l and the line $2x - 3y = 4$, leaving the answer in exact form. [2]

14. It is given that $a * b = a + b + 1$ is a binary operation on the set of integers $\mathbb{Z}$.

    (a) Find $-2 * 7$. [1]

    (b) Show that the set of integers $(\mathbb{Z})$ under the binary operation $a * b = a + b + 1$ forms an Abelian group. [8]

15. The functions f and g are defined by:

$$f : x \to 3x + p, \quad g : x \to q - 2x; \quad x \in \mathbb{R},$$

where p and q are constants. Given that $ff(2) = 10$ and $g^{-1}(2) = 3$, find:

    (a) the value of p and the value of q. [7]

    (b) an expression for $fg(x)$ using the values of p and q from (a). [2]

16. Prove by induction that:

$$\frac{5^{2n} + 5}{5},$$

is divisible by 3, for all integer values of n. [10]

17. (a) Express:

$$f(x) = \frac{3x + 14}{(x-2)(x^2+6)},$$

in the form:

$$\frac{A}{x-2} + \frac{Bx + C}{x^2 + 6}.$$

[5]

    (b) Hence or otherwise, find the exact value of:

$$\int_3^4 f(x) \ \mathbf{d}x.$$

[6]

QUESTIONS

Question 1

$$
\begin{aligned}
\frac{3-x}{3+2x} &= (3-x)(3+2x)^{-1} \\
&= (3-x)\cdot 3^{-1}\left(1+\frac{2}{3}x\right)^{-1} \\
&= \frac{1}{3}(3-x)\left[1+(-1)\left(\frac{2}{3}x\right)+\frac{(-1)(-2)}{2!}\left(\frac{2}{3}x\right)^2\right. \\
&\quad \left.+\frac{(-1)(-2)(-3)}{3!}\left(\frac{2}{3}x\right)^3+\cdots\right] \\
&= \left(1-\frac{x}{3}\right)\left(1-\frac{2}{3}x+\frac{4}{9}x^2-\frac{8}{27}x^3+\cdots\right) \\
&= 1-\frac{2}{3}x+\frac{4}{9}x^2-\frac{8}{27}x^3-\frac{x}{3}+\frac{2}{9}x^2-\frac{4}{27}x^3+\cdots \\
&= 1-x+\frac{2}{3}x^2-\frac{4}{9}x^3+\cdots
\end{aligned}
$$

For validity;

$$-\frac{3}{2}<x<\frac{3}{2}$$

Question 2

$$
\begin{aligned}
\frac{3^x(9^x)^2}{81} &= 27^x \\
\frac{3^x(3^{2x})^2}{3^4} &= 3^{3x} \\
\frac{3^{x+4x}}{3^4} &= 3^{3x} \\
5x-4 &= 3x \\
5x-3x &= 4 \\
2x &= 4 \\
x &= 2
\end{aligned}
$$

Question 3

(a)

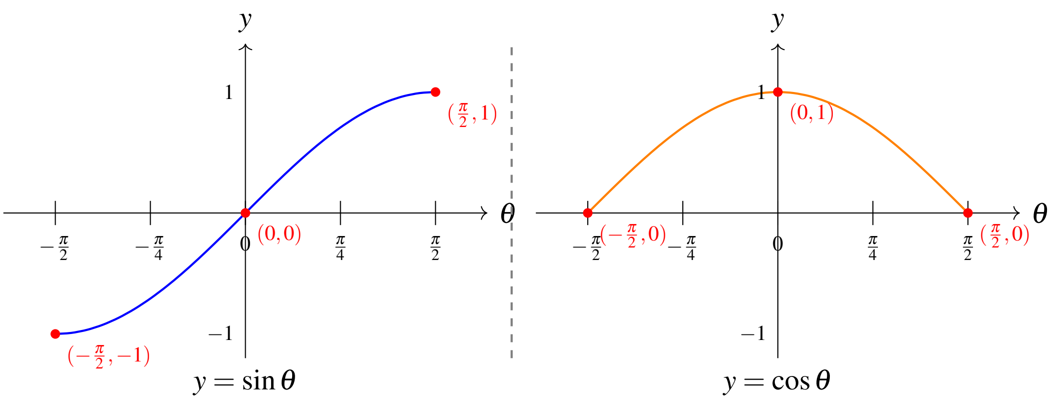
$$x = \frac{\sin\theta}{\cos\theta} \quad (\tan\theta)$$
$$\frac{dx}{d\theta} = \frac{\cos\theta\cdot\cos\theta - \sin\theta(-\sin\theta)}{\cos^2\theta}$$
$$= \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta}$$
$$= \frac{1}{\cos^2\theta} = \sec^2\theta$$

$$y = 1 - \sin^2\theta \quad (\text{since } \cos^2\theta = 1-\sin^2\theta)$$
$$= \cos^2\theta$$
$$\frac{dy}{d\theta} = 2\cos\theta\cdot(-\sin\theta)$$
$$= -2\sin\theta\cos\theta$$
$$= -\sin 2\theta$$

$$\therefore \frac{dy}{dx} = \frac{-2\sin\theta\cos\theta}{\sec^2\theta}$$
$$= -2\sin\theta\cos^3\theta \quad \text{or} \quad \frac{-\sin 2\theta}{\sec^2\theta}$$

(b) For critical points $(dy/dx = 0)$

$$-2\sin\theta\cos^3\theta = 0$$
$$\Rightarrow -2\sin\theta = 0 \quad \text{or} \quad \cos^3\theta = 0$$
$$\sin\theta = 0 \quad \text{or} \quad \cos\theta = 0$$
$$\theta = 0 \quad \text{or} \quad \theta = \pm\frac{\pi}{2} \quad \textit{Note: Ignore since } -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$



If $\theta = 0$:

$$x = \tan(0) = 0$$
$$y = (\cos 0)^2 = 1$$

Therefore the coordinates $= (0; 1)$

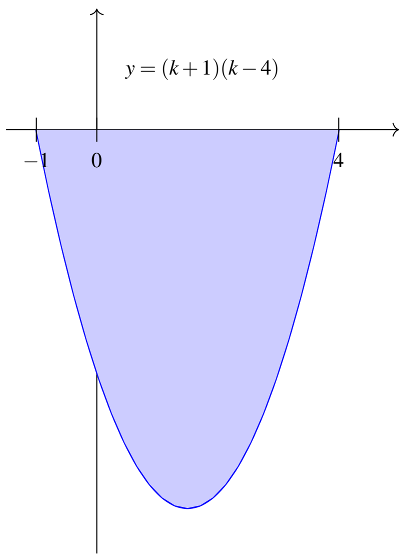
Question 4

$$\text{Equation: } kx^2 + 4x + k = 3$$
$$kx^2 + 4x + (k-3) = 0$$

For real roots ($b^2 - 4ac \geq 0$):

$$4^2 - 4(k)(k-3) \geq 0$$
$$16 - 4k^2 + 12k \geq 0$$
$$4k^2 - 12k - 16 \leq 0$$
$$k^2 - 3k - 4 \leq 0$$
$$(k-4)(k+1) \leq 0$$

Critical values: $k = 4$ or $k = -1$



$$\therefore -1 \leq k \leq 4$$

Question 5

(a)

$$y = x^m e^n$$
$$\ln y = \ln(x^m e^n)$$
$$\ln y = \ln x^m + \ln e^n$$
$$\ln y = m \ln x + n$$

In standard linear form:

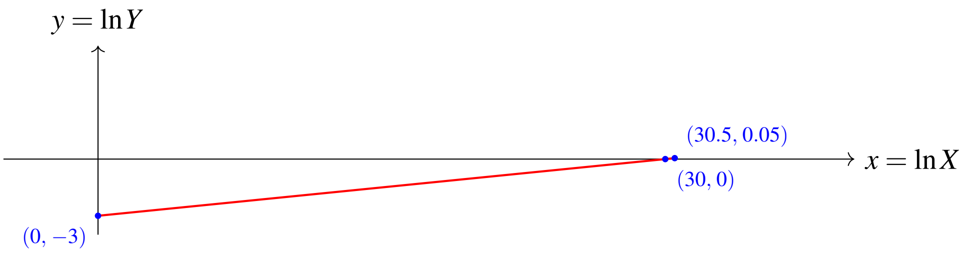
$$\ln y = m \ln x + n$$

which corresponds to:

$$Y = mX + c$$

where $Y = \ln y$, $X = \ln x$, and $c = n$.

(b) Sketch



$$m = \frac{-3 - 0.05}{0 - 30.5}$$
$$m = \frac{1}{10} = 0.1$$

Given:

$$\text{If } \ln x = 0, \text{ then } \ln y = -3 \quad (\text{In } y \text{ - intercept})$$
$$\Rightarrow n = -3$$

Question 6

(a) Given function:

$$f(x) = (x-2)(x+3) + ax + b$$

Conditions:

$$\begin{aligned} f(2) &= 8 \\ (2-2)(2+3)+2a+b &= 8 \\ b &= 8-2a \quad \text{(Equation 1)} \end{aligned}$$

$$\begin{aligned} f(-3) &= -12 \\ (-3-2)(-3+3)+(-3)a+b &= -12 \\ (-5)(0)-3a+b &= -12 \\ b &= 3a-12 \quad \text{(Equation 2)} \end{aligned}$$

Solving the equations:

$$\begin{aligned} \text{From Equation 1: } & b = 8-2a \\ \text{Substitute into Equation 2: } & 8-2a = 3a-12 \\ & 20 = 5a \\ & a = 4 \\ \text{Then from equation 1: } & b = 8-2(4) = 0 \\ \text{Finally: } & a = 4 \quad \text{and} \quad b = 0 \end{aligned}$$

(b)

$$\frac{(x-2)(x+3)+4x}{(x-2)(x+3)} = 1+\frac{4x}{(x-2)(x+3)} \quad \left(\text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}}\right)$$

$$\therefore \text{Remainder} = 4x$$

Question 7

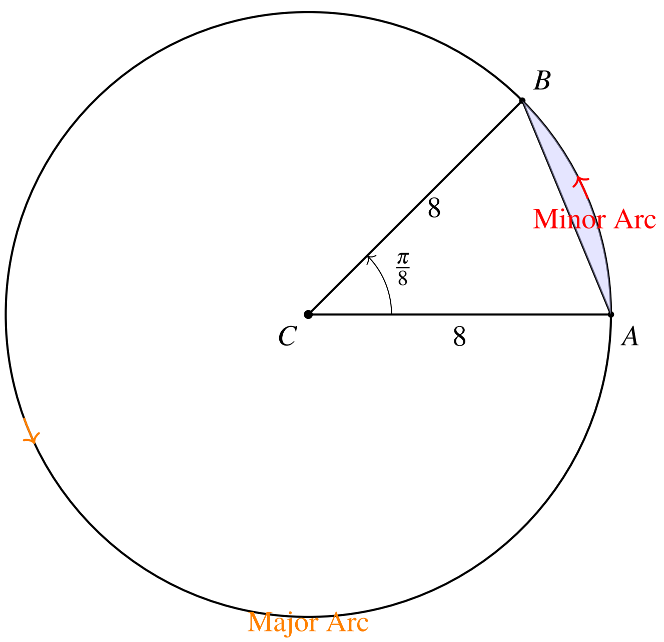
$$x\frac{\mathbf{d}y}{\mathbf{d}x} = y+1$$
$$\int \frac{1}{y+1} \quad \mathbf{d}y = \int \frac{1}{x} \quad \mathbf{d}x$$
$$\ln(y+1) = \ln x + c \quad \text{[General Solution]}$$

$$\text{If } y = 4; \quad x = 2$$

$$\ln(4+1) = \ln 2 + c$$
$$c = \ln\left(\frac{5}{2}\right)$$
$$\therefore \ln(y+1) = \ln x + \ln\left(\frac{5}{2}\right)$$
$$\cancel{\ln}(y+1) = \cancel{\ln}\left(\frac{5x}{2}\right)$$
$$(y+1) = \left(\frac{5x}{2}\right)$$
$$\therefore y = \frac{5}{2}x - 1 \quad \text{[Particular Solution or } (y = 2.5x-1)\text{]}$$

Question 8

(a) Length of the major arc l:



$$\begin{aligned} l &= r\theta \\ &= 8\left(2\pi - \frac{\pi}{8}\right) \\ &= 15\pi \text{ cm} \end{aligned}$$

(b) Area of the minor segment A;

$$\begin{aligned} A &= \frac{1}{2}r^2(\theta - \sin\theta) \\ &= \frac{1}{2}(8)^2\left[\frac{\pi}{8} - \sin\frac{\pi}{8}\right] \\ &= 4\pi - 32\sin\frac{\pi}{8} \\ &= 0.32 \text{ cm}^2 \quad [2 \text{ s.f}] \end{aligned}$$

Question 9

(a)

$$\begin{aligned} PQ &= \begin{pmatrix} -1 & 1 & 3 \\ 2 & -2 & -6 \\ -4 & 4 & 12 \end{pmatrix}\begin{pmatrix} 3 & -2 & 0 \\ -1 & 3 & 7 \\ 2 & -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -3-1+6 & 2+3-3 & 0+7+3 \\ 6+2-12 & -4-6+6 & 0-14-6 \\ -12-4+24 & 8+12-12 & 0+28+12 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 2 & 10 \\ -4 & -4 & -20 \\ 8 & 8 & 40 \end{pmatrix} \end{aligned}$$

(b) If a matrix does not have an inverse then: $\det = 0$ hence singular or non invertible matrix

$$\begin{aligned} \det &= -6\begin{vmatrix} -6 & 0 \\ 10 & -4 \end{vmatrix} + 2\begin{vmatrix} 4 & 0 \\ 8 & -4 \end{vmatrix} + 2\begin{vmatrix} 4 & -6 \\ 8 & 10 \end{vmatrix} \\ &= -6(24) + 2(-16) + 2(88) \\ &= -144 - 32 + 176 \\ &= 0 \quad \text{(Shown)} \end{aligned}$$

Question 10

(a)

$$
\begin{aligned}
F &\alpha W^2 \\
F &= kW^2 \\
165 &= k(830)^2 \\
165 &= 688\,900k \\
k &= \frac{33}{137\,780} \\
F &= \frac{33}{137\,780}W^2
\end{aligned}
$$

(b) If $F = 32$;

$$
\begin{aligned}
132 &= \frac{33}{137\,780}W^2 \\
W^2 &= \frac{132(137\,780)}{33} \\
W &= \pm\sqrt{551\,120} \\
W &= 742.3745685\ldots \\
W &= \pm 740 \quad [2 \text{ s.f}]
\end{aligned}
$$

Question 11

(a)

$$
\begin{aligned}
e^{-1/2x} &= 3x - 1 \\
\text{Let } f(x) &= e^{-1/2x} - 3x + 1
\end{aligned}
$$

Then;

$$
\begin{aligned}
f(0) &= e^{(0)} - 3(0) + 1 = 2 > 0 \\
f(1) &= e^{-1/2} - 3 + 1 \\
&= -1.393\cdots < 0
\end{aligned}
$$

Hence, change of sign indicates root between 0 and 1.

(b)

$$\begin{aligned} e^{-\frac{1}{2}x} &= 3x-1 \\ -\frac{1}{2}x &= \ln(3x-1) \\ x &= -2\ln(3x-1) \end{aligned}$$

Putting subscripts;

$$x_{n+1} = -2\ln(3x_n - 1) \quad \text{[Shown]}$$

(c)

$$\begin{aligned} x_1 &= 0.5 \\ x_2 &= -2\ln[3(0.5)-1] \\ &= -2\ln(0.5) \\ &= 1.386\ldots \\ &= 1.386 \quad \text{[3 d.p]} \end{aligned}$$

Question 12

$$\sqrt[3]{1-\sqrt{3}i} = \left(1-\sqrt{3}i\right)^{1/3}$$

$$\text{Let } z = 1-\sqrt{3}i$$

$$r = \sqrt{1^2+(-\sqrt{3})^2} = 2$$

$$\theta = \tan^{-1}\left(-\sqrt{3}\right) = -\frac{\pi}{3}$$

$$z = 2\left[\cos\left(-\frac{\pi}{3}\right)+i\sin\left(-\frac{\pi}{3}\right)\right]$$

$$\therefore z^{\frac{1}{3}} = 2^{1/3}\left[\cos\left(-\frac{\pi}{3}+2k\pi\right)+i\sin\left(-\frac{\pi}{3}+2k\pi\right)\right]$$

$$= \sqrt[3]{2}\left[\cos\left(-\frac{\pi+6k\pi}{9}\right)+i\sin\left(-\frac{\pi+6k\pi}{9}\right)\right]$$

For $k=0$

$$z_1 = \sqrt[3]{2}\left[\cos\left(-\frac{\pi}{9}\right)+i\sin\left(-\frac{\pi}{9}\right)\right]$$

$$= 1,183938513\cdots - 0,4309\ldots i$$

$$= 1,2-0.43i \quad [2 \text{ s.f}]$$

For $k=1$

$$z_2 = \sqrt[3]{2}\left[\cos\left(\frac{5\pi}{9}\right)+i\sin\left(\frac{5\pi}{9}\right)\right]$$

$$= -0,218782\cdots + 1,24078\ldots i$$

$$= -0.22+1.2i \quad [2 \text{ s.f}]$$

For $k=-1$

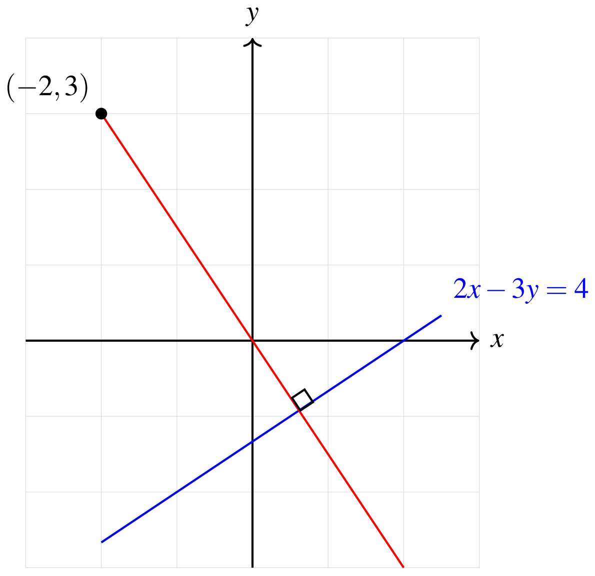
$$z_3 = \sqrt[3]{2}\left[\cos\left(-\frac{7\pi}{9}\right)+i\sin\left(-\frac{7\pi}{9}\right)\right]$$

$$= -0,9651555\cdots - 0,809821640\ldots i$$

$$= 0.97-0.81i \quad [2 \text{ s.f}]$$

Question 13

(a) Sketch



$$2x - 3y = 4 \quad (1) \tag{1}$$
$$3y = 2x - 4 \tag{2}$$
$$y = \frac{2}{3}x - \frac{4}{3} \tag{3}$$

Slope (Gradient, m) of line l:

$$m = \frac{-1}{\frac{2}{3}} = \frac{-3}{2}$$

Equation of line l:

$$y - 3 = -\frac{3}{2}(x+2) \tag{4}$$
$$y = -\frac{3}{2}x - 3 + 3 \tag{5}$$
$$y = -\frac{3}{2}x \quad (2) \tag{6}$$

Substitute equation (2) into (1):

$$2x - 3\left(\frac{-3}{2}x\right) = 4 \tag{7}$$
$$2x + \frac{9x}{2} = 4 \tag{8}$$
$$\frac{13x}{2} = 4 \tag{9}$$
$$x = \frac{4 \times 2}{13} = \frac{8}{13} \tag{10}$$

From equation (2):

$$y = -\frac{3}{2}\left(\frac{8}{13}\right) = \frac{-12}{13}$$

$$\therefore \text{ Coordinates} = \left(\frac{8}{13}, -\frac{12}{13}\right) \quad \text{(Point P)}$$

(b) Distance, d

$$\begin{aligned} d &= \sqrt{\left(-2-\frac{8}{13}\right)^2+\left(3+\frac{12}{13}\right)^2} \\ &= \sqrt{\frac{289}{13}} \\ &= \frac{17\sqrt{13}}{13} \quad \text{units (exact form)} \end{aligned}$$

Question 14

(a)

$$\begin{aligned} -2*7 &= -2+7+1 \\ &= 6 \end{aligned}$$

(b) Closure property

Let $a, b \in \mathbb{Z}$. Then,

$$a*b = a+b+1 \in \mathbb{Z}$$

Since integers are closed under addition and 1 is an integer, $a*b \in \mathbb{Z}$. Thus, closure holds true.

Associativity

$$\begin{aligned} (a*b)*c &= (a+b+1)*c \\ &= a+b+1+c+1 \\ &= a+b+c+2 \end{aligned}$$

$$
\begin{aligned}
a*(b*c) &= a*(b+c+1) \\
&= a+b+c+1+1 \\
&= a+b+c+2
\end{aligned}
$$

Associativity holds true since $(a*b)*c = a*(b*c)$.

Identity Element

We seek an element $e \in \mathbb{Z}$ such that for any $a \in \mathbb{Z}$:

$$a*e = a \quad \text{and} \quad e*a = a$$

So:

$$
\begin{aligned}
a*e &= a+e+1 = a \\
\Rightarrow e+1 &= 0 \\
\Rightarrow e &= -1
\end{aligned}
$$

Similarly:

$$
\begin{aligned}
e*a &= e+a+1 = a \\
\Rightarrow e+1 &= 0 \\
\Rightarrow e &= -1
\end{aligned}
$$

Thus, $e = -1$ is the identity element in $\mathbb{Z}$, hence identity exists.

Inverse Element

For each $a \in \mathbb{Z}$, we seek an element $b \in \mathbb{Z}$ such that:

$$
\begin{aligned}
a*b &= e = -1 \\
\Rightarrow a+b+1 &= -1 \\
\Rightarrow b &= -2-a \\
\therefore a^{-1} &= -a-2 \\
&= b \in \mathbb{Z}
\end{aligned}
$$

Hence, inverse exists.

Commutativity:

$$a * b = a + b + 1$$
$$b * a = b + a + 1 = a + b + 1$$

Since $a * b = b * a$ hence abelian.

The set of integers $\mathbb{Z}$ with the binary operation $a * b = a + b + 1$ forms an **Abelian group**.

Question 15

(a)

$$\begin{aligned} f(f(2)) &= 10 \quad \text{(Compound function)} \\ f[3(2)+p] &= 10 \\ f(6+p) &= 10 \\ 3(6+p)+p &= 10 \\ 18+3p+p &= 10 \\ 4p &= -8 \\ p &= -2 \end{aligned}$$

Let $y = g(x)$:

$$\begin{aligned} y &= q - 2x \\ \text{Solve for } x: & \\ x &= \frac{q-y}{2} \\ \text{Thus, the inverse function is:} & \\ g^{-1}(x) &= \frac{q-x}{2} \end{aligned}$$

Given $g^{-1}(2) = 3$:

$$\begin{aligned} \frac{q-2}{2} &= 3 \\ q-2 &= 6 \\ q &= 8 \end{aligned}$$

(b) Composite function $fg(x)$:

$$\begin{aligned} fg(x) &= f(8-2x) \\ &= 3(8-2x) + (-2) \\ &= 24 - 6x - 2 \\ &= 22 - 6x \end{aligned}$$

Question 16

Let P_n be the statement:

$$\frac{5^{2n}+5}{5} \text{ is divisible by } 3$$

Base Case ($n = 1$):

$$\begin{aligned} \frac{5^{2(1)}+5}{5} &= \frac{25+5}{5} \\ &= \frac{30}{5} \\ &= 6 \quad \text{(which is divisible by 3 and also a multiple of 3)} \end{aligned}$$

Thus, P_1 is true.

Inductive Hypothesis: Assume P_k is true for some $n = k$, i.e.:

$$\frac{5^{2k}+5}{5} = 3d \quad \text{for some integer } d$$

which implies:

$$5^{2k} + 5 = 15d$$

$$5^{2k} = 15d - 5$$

Inductive Step ($n = k+1$):

$$\begin{aligned} \frac{5^{2(k+1)}+5}{5} &= \frac{5^{2k+2}+5}{5} \\ &= \frac{25 \cdot 5^{2k}+5}{5} \\ &= \frac{25(15d-5)+5}{5} \quad \text{(by inductive hypothesis)} \\ &= \frac{375d-125+5}{5} \\ &= 75d - 24 \\ &= 3(25d-8) \quad \text{(Hence, } P_{k+1} \text{ is also true)} \end{aligned}$$

Since P_1, P_k and P_{k+1} hold true, hence by mathematical induction P_n is true, $\forall\, n \in \mathbb{Z}^+$.

Question 17

(a) Given:

$$f(x) = \frac{3x+14}{(x-2)(x^2+6)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+6}$$

Now, solving for coefficients A, B, and C:

If $x = 2$:

$$A = \frac{3(2)+14}{2^2+6} = \frac{20}{10} = 2$$

If $x = 0$:

$$\frac{14}{(-2)(6)} = \frac{A}{-2} + \frac{C}{6} \Rightarrow -\frac{7}{6} = -\frac{2}{2} + \frac{C}{6}$$

$$-\frac{7}{6} = -1 + \frac{C}{6} \Rightarrow \frac{C}{6} = \frac{-1}{6} \Rightarrow C = -1$$

If $x = 1$:

$$\frac{17}{(-1)(7)} = \frac{2}{-1} + \frac{B(1)-1}{7} \Rightarrow -\frac{17}{7} = -2 + \frac{B-1}{7}$$

$$-\frac{17}{7} + 2 = \frac{B-1}{7} \Rightarrow -\frac{3}{7} = \frac{B-1}{7} \Rightarrow B = -2$$

Thus, the partial fractions decomposition is:

$$f(x) = \frac{2}{x-2} + \frac{-2x-1}{x^2+6}$$

Alternative form:

$$f(x) = \frac{2}{x-2} - \frac{2x+1}{x^2+6}$$

(b)

$$\int_3^4 \left(\frac{3x+14}{(x-2)(x^2+6)}\right) \mathbf{d}x = \int_3^4 \left(\frac{2}{x-2} + \frac{-2x-1}{x^2+6}\right) \mathbf{d}x$$

$$\int_3^4 \left(\frac{2}{x-2} + \frac{-2x-1}{x^2+6}\right) \mathbf{d}x = \int_3^4 \left(\frac{2}{x-2} + \frac{-2x}{x^2+6} + \frac{-1}{x^2+6}\right) \mathbf{d}x$$

$$= \int_3^4 \left(\frac{2}{x-2} + \frac{-2x}{x^2+6} - \frac{1}{x^2+(\sqrt{6})^2}\right) \mathbf{d}x$$

$$= \left[2\ln|x-2| - \ln|x^2+6| - \frac{1}{\sqrt{6}}\tan^{-1}\left(\frac{x}{\sqrt{6}}\right)\right]_3^4$$

$$= \left[2\ln 2 - \ln 22 - \frac{1}{\sqrt{6}}\tan^{-1}\left(\frac{4}{\sqrt{6}}\right)\right] - \left[-\ln 15 - \frac{1}{\sqrt{6}}\tan^{-1}\left(\frac{3}{\sqrt{6}}\right)\right]$$

$$= \ln 4 - \ln 22 - \frac{1}{\sqrt{6}}\tan^{-1}\left(\frac{2\sqrt{6}}{3}\right) + \ln 15 + \frac{1}{\sqrt{6}}\tan^{-1}\left(\frac{\sqrt{6}}{2}\right)$$

$$= \ln\left(\frac{4}{22}\times 15\right) + \frac{1}{\sqrt{6}}\left[\tan^{-1}\left(\frac{\sqrt{6}}{2}\right) - \tan^{-1}\left(\frac{2\sqrt{6}}{3}\right)\right]$$

$$= \ln\left(\frac{30}{11}\right) + \frac{\sqrt{6}}{6}\left[\tan^{-1}\left(\frac{\sqrt{6}}{2}\right) - \tan^{-1}\left(\frac{2\sqrt{6}}{3}\right)\right] \quad \text{(exact form)}$$

Standard Integral Formulas Used

$$\int \frac{f'(x)}{f(x)} \mathbf{d}x = \ln|f(x)| + C$$

$$\int \frac{1}{a^2+x^2} \mathbf{d}x = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$$

END OF SOLUTIONS

FEEL FREE TO CONTACT US FOR ANY ADJUSTMENTS, CLARIFICATIONS AND ASSISTANCE!

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